LostLink: A Review of Intelligent Lost-and-Found System

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Abstract. Lost-and-found recovery on campuses and in public spaces still depends largely on notice boards, informal messaging groups, and keyword-driven listing portals. Finding the one report that matches a given item is slow and unreliable under this model, and ownership is essentially unverified once a possible match is found. This paper surveys the techniques available for both halves of the problem, automatic item matching and evidence-based ownership verification, and proposes a methodology that addresses the gaps the survey identifies. On the matching side, we trace the shift from keyword and category search, through convolutional Siamese networks fine-tuned on category-specific image pairs, to zero-shot vision embedding models such as CLIP and DINOv2 that generalise across object types without retraining, and to approaches that fuse image similarity with text, location, and time signals. On the verification side, we examine how existing lost-and-found prototypes handle ownership claims and contrast this with the false-acceptance/false-rejection framework used more broadly to evaluate authentication systems. The review finds that most matching systems in this space remain tied to a single fine-tuned object category, and that ownership verification is either absent, reduced to a login check, or described only conceptually, without measured accuracy. We then present the proposed methodology for our own system, LostLink, which evaluates zero-shot embeddings across multiple item categories and introduces a scored, evidence-based verification mechanism assessed using standard authentication metrics.

Keywords: lost and found · image matching · CLIP · DINOv2 · multimodal fusion · ownership verification · Siamese networks

1 Introduction

Misplacing personal belongings is an everyday occurrence on college campuses, in transit hubs, and in offices. Recovering them typically comes down to physical notice boards, informal WhatsApp or Instagram groups, or simply asking around. A number of digital lost-and-found portals now exist, but most of them

still organise reports by keyword or category, which helps only a user who happens to browse the correct entry. Picking the one genuine match out of many visually similar listings remains largely a matter of luck. Even when a plausible match is located, there is usually no structured way to confirm that the person claiming an item is its actual owner, so recovery still relies on administrator judgement rather than evidence.

Two research threads are relevant to closing this gap. The first is automatic item matching, which has moved from manual keyword search toward computer-vision techniques capable of comparing photographs directly. The second is ownership verification, an authentication problem that lost-and-found systems have so far addressed only informally, if at all. This paper reviews both threads and then proposes a methodology, for a system we call LostLink, that is designed around the gaps the review surfaces.

The remainder of the paper is organised as follows. Section 2 is the literature review: it covers keyword- and metadata-based matching as currently deployed (Section 2.1), deep-learning approaches to image matching, from fine-tuned convolutional Siamese networks to zero-shot embedding models (Section 2.2), multimodal fusion of image, text, location, and time signals (Section 2.3), and ownership verification together with the metrics used to evaluate authentication systems generally (Section 2.4), before summarising all of the above in a comparative table (Section 2.5). Section 3 discusses the two gaps this review identifies. Section 4 presents the proposed methodology for LostLink. Section 5 concludes.

2 Literature Review

2.1 Keyword- and Metadata-Based Matching

Most deployed lost-and-found portals, including typical campus systems, are built around user registration, a keyword or category search form, and an administrator panel that manually reviews reports [3, 15]. UniFind, for example, layers natural-language processing and basic image recognition on top of this structure to suggest candidate matches, but verifies claimants through their college login credentials rather than through any evidence tied to the item itself [15]. A related class of system pairs registration with QR-code tagging, so that a found item scanned by a finder is linked directly back to its registered owner's profile. This sidesteps matching altogether for pre-registered items, but offers no help for items that were never tagged [3]. Zhao and Peng describe a lost-and-found system built around a map API that adds location context to listings, though matching in that system is still driven by manually entered fields rather than automated similarity [18], and Xue et al. propose recording lost-and-found transactions on a blockchain primarily to make the audit trail tamper-resistant, again leaving the underlying matching step to keyword search [16].

The shared limitation across this category of system is that two people rarely describe the same object in the same words. Keyword matching therefore misses genuine matches whenever the finder's and the owner's descriptions diverge, and

it makes no use of the signal that is usually most distinctive: what the item actually looks like.

2.2 Deep-Learning-Based Image Matching

Image-based matching replaces manual description matching with a direct comparison of photographs. The systems reviewed here fall into two families: convolutional networks fine-tuned for a specific object category, and pretrained embedding models used zero-shot.

Fine-Tuned Convolutional and Siamese Matching. The Siamese-network idea, training twin networks with shared weights to output a similarity score for a pair of images rather than a fixed class label, was introduced by Koch et al. for one-shot image recognition and underlies most category-specific lost-and-found matchers [10]. Dhanawardhana et al. apply this idea directly to lost-and-found recovery, fine-tuning a MobileNetV2-based [14] Siamese network on 848 hand-labelled pairs of bag photographs and fusing the resulting visual-similarity score with text, location, and time similarity into a single ranked score [8]. Zhou et al.'s LostNet takes a related but image-only route, combining a MobileNetV2 backbone with a Convolutional Block Attention Module and a perceptual-hashing step for fast approximate matching, reporting 96.8% test accuracy at low computational cost [19]. A separate study applies ResNet-50 with a distance-based similarity formula to the same problem and reports only around 30% matching accuracy, which illustrates that a classification-oriented backbone does not automatically transfer well to open-set similarity matching [12].

The common constraint across this family is that a Siamese network's usefulness depends entirely on the pairs it was fine-tuned on. A model tuned on bag photographs, as in Dhanawardhana et al., has no established ability to generalise to backpacks, phones, keys, or umbrellas without a fresh labelled dataset being collected for each new category [8], which is impractical for a general-purpose lost-and-found platform expected to handle arbitrary item types.

Zero-Shot Vision-Language and Self-Supervised Embeddings. A separate line of pretrained embedding models sidesteps category-specific fine-tuning by treating visual similarity as the direct training objective rather than a side-effect of classification. CLIP is trained contrastively on large-scale image-caption pairs so that semantically related images and text land close together in a shared embedding space, and it performs competitively on classification and retrieval tasks with no task-specific fine-tuning [13]. DINOv2 instead uses self-supervised distillation over a large curated image corpus, producing general-purpose visual features without requiring any labels or paired captions at all [11]. Both models are increasingly used off-the-shelf for open-set re-identification, but the two are not interchangeable. A recent comparison of CLIP, DINOv2, DINOv3, and other pretrained encoders on zero-shot object re-identification finds that self-supervised, purely visual encoders such as DINOv2 tend to separate visually

similar instances of the same object category more cleanly than the language-supervised CLIP encoder, which is comparatively stronger when textual cues, rather than fine-grained visual detail, drive the match [6]. A broader survey of transformer-based re-identification reaches a similar conclusion from a different angle, noting that CLIP is a common starting point for re-identification work chiefly because of its strong zero-shot generalisation across domains, rather than because it is the sharpest available instance-level matcher [17].

A direct data point on CLIP's usefulness for this exact application comes from an institutional lost-and-found prototype that adopts OpenAI's CLIP (ViT-L/14) as its matching backbone and evaluates it on a 500-image subset of CIFAR-10, reporting 95.20% accuracy, 95.43% precision, 95.35% recall, and a 95.28% F1-score for matching images to their text descriptions [7]. This is a substantially stronger result than the fine-tuned CNN baselines discussed above, where the ResNet-50 similarity model reaches only 29.96% image-matching accuracy [12] and even Dhanawardhana et al.'s purpose-trained Siamese network reaches only 0.75 accuracy against 0.9526 for its text-based SBERT component [8]. The comparison is nonetheless not quite apples-to-apples: CIFAR-10 is a generic ten-class object benchmark rather than a lost-and-found dataset, so this result demonstrates that CLIP embeddings can support high-confidence image-text matching in principle, without yet showing how the same embeddings perform on the near-duplicate, single-category confusions, for instance two black backpacks or two identical phone cases, that make lost-and-found matching hard in practice. No system reviewed here reports a head-to-head comparison of CLIP against DINOv2 on lost-and-found imagery specifically.

Taken together, the evidence supports a specific, testable prediction rather than a settled answer. DINOv2-style self-supervised embeddings should be expected to outperform CLIP on the fine-grained, single-category visual matching that dominates lost-and-found reports, based on the general re-identification evidence in [17] and [6], while CLIP's joint image-text space may still be the more natural choice wherever a textual description needs to be matched directly against an image, or where no photograph is available at all. For a lost-and-found platform, the practical implication is that CLIP- or DINOv2-style embeddings can, in principle, compare a lost backpack against a found backpack, and a lost phone against a found phone, using the same model and no per-category retraining, removing the fine-tuning bottleneck identified above. Whether this holds across a wide range of everyday object categories at the accuracy level CLIP achieved on CIFAR-10 [7], rather than only on narrow benchmarks, is the open question this review surfaces.

2.3 Multimodal Fusion of Image, Text, Location, and Time

Image similarity alone is not sufficient: two different black backpacks can look nearly identical, while the correct match is often disambiguated by where and when an item was lost or found. The general problem of aligning image and text representations for cross-modal retrieval has been studied extensively; Khandelwal and Poria survey the main strategies for combining embeddings from sepa-

rate modalities into a joint similarity score [9], and more recent work such as the Progressive Multi-Subspace Fusion model shows that fusing features from multiple encoder layers, rather than only the final embedding, improves cross-modal retrieval accuracy [4]. Within lost-and-found specifically, Dhanawardhana et al.'s system is the clearest precedent for this kind of fusion, computing a weighted sum of image, text, location, and time similarity as the final match score [8], a design choice several subsequent lost-and-found prototypes describe adopting in principle, without publishing a comparable evaluation of the fused score against its individual components [7].

What is largely missing from this literature, as distinct from general text-image retrieval research, is a systematic comparison of how much each modality, image, text, location, and time, actually contributes to correct-match accuracy in the lost-and-found setting specifically, and whether that contribution changes across object categories.

2.4 Ownership Verification and Claim Arbitration

Verification in Existing Lost-and-Found Systems. Once a candidate match is surfaced, most existing prototypes stop short of verifying that the claimant is the true owner. UniFind verifies claimants through college-ID login and account credentials rather than through evidence connected to the specific item, which establishes who is asking but not whether they own the item in question [15]. A 2026 preprint on campus claim arbitration goes further, describing a workflow for resolving disputes when multiple people claim the same found item, but it is not peer-reviewed and reports no quantitative accuracy for how well the arbitration mechanism separates genuine owners from false claimants [5]. A broader survey of university lost-and-found practice finds that a large share of lost items on campus are simply never recovered, and attributes this in part to the absence of a reliable, low-friction way to confirm ownership at the point of return [3]. The CLIP-based institutional prototype discussed in Section 2.2 reports strong image-matching accuracy but, like the others, includes only an administrator-verification step, with no scored, evidence-based ownership check and no reported accuracy for that step [7].

Verification as an Authentication Problem. Outside the lost-and-found literature, verifying a claimed identity from evidence the claimant supplies is a well-studied authentication problem, typically evaluated with two competing error rates: the False Acceptance Rate (FAR), the proportion of illegitimate claimants who are wrongly approved, and the False Rejection Rate (FRR), the proportion of genuine claimants who are wrongly denied [1]. Because tightening a verification threshold to reduce FAR generally raises FRR and vice versa, such systems are usually evaluated across a range of thresholds rather than at a single operating point [1]. The EVI benchmark applies this framework to spoken knowledge-based verification, where a system asks a claimant for private details and scores their answers, and reports FRR at a fixed FAR alongside an Equal

Error Rate at which the two error types are balanced [2]. Multi-claimant identification, where several people compete to be recognised as the correct owner, is correspondingly evaluated by identification-rate-at-rank, the proportion of cases in which the correct identity appears among the top-ranked candidates rather than strictly first [2].

None of the lost-and-found systems reviewed in the previous section apply this FAR/FRR framework, or any comparable quantitative measure, to their ownership-verification step. This is the second gap this review identifies: a scored, evidence-based verification mechanism for lost-and-found claims, evaluated the way an authentication system normally would be, across thresholds, with both error rates reported, does not appear to exist in the published lost-and-found literature.

2.5 Comparative Summary

Table 1 summarises the matching and verification approaches reviewed above.

Table 1. Comparative summary of matching and verification approaches.

Category	Representative work	Strengths	Limitations
Keyword / metadata search	Campus portals; UniFind [15]; map-based system [18]	Simple to build; low compute cost	Misses matches when descriptions differ in wording
Fine-tuned CNN / Siamese matching	Dhanawardhana et al. [8]; LostNet [19]; ResNet-50 similarity [12]	High accuracy within the trained category; lightweight backbones	Tied to one object category; needs fresh labelled pairs to extend
Zero-shot embeddings (CLIP / DINOv2)	Radford et al. [13]; Oquab et al. [11]; CLIP on CIFAR-10 [7]; re-ID comparisons [17, 6]	No per-category fine-tuning; generalises across item types	No head-to-head CLIP vs. DINOv2 test on lost-and-found imagery specifically
Multimodal fusion	Dhanawardhana et al. [8]; PMSF [4]; survey [9]	Uses location/time context that image alone ignores	Per-modality contribution rarely measured in lost-and-found settings
Ownership verification	Login-based checks [15]; claim-arbitration preprint [5]; FAR/FRR framework [1, 2]	FAR/FRR gives a standard, threshold-based evaluation	Not applied, with reported accuracy, to lost-and-found ownership claims

3 Research Gaps

Two gaps stand out from this review. First, the matching systems that report strong accuracy [8, 19] achieve it by fine-tuning on a single object category. A CLIP-based prototype independently reports 85.2% accuracy [7], but that figure comes from a generic ten-class benchmark rather than the near-duplicate, single-category confusions typical of real lost-and-found reports. Whether zero-shot embedding models, and specifically DINOv2, which the broader re-identification literature suggests should outperform CLIP on exactly this kind of fine-grained visual matching [17, 6], can match or exceed fine-tuned-CNN accuracy across a broad range of item categories without any fine-tuning at all remains an open, testable question.

Second, ownership verification in lost-and-found systems is currently either absent, reduced to a login or ID check, or described only as a workflow without measured performance [15, 5], while a standard, threshold-based evaluation framework for exactly this kind of evidence-based verification already exists in the authentication literature and has simply not been applied here [1, 2]. Addressing both gaps with measurable, cross-category results, including a direct CLIP-versus-DINOv2 comparison on lost-and-found imagery, is the direction the proposed methodology below pursues.

4 Proposed Methodology

This section describes the system being built to address the two gaps identified in Section 3: LostLink, a lost-and-found platform that evaluates zero-shot embeddings across item categories and pairs matching with a quantitatively evaluated ownership-verification mechanism.

4.1 System Overview

LostLink accepts a report for a lost or a found item consisting of a photograph, an item category, a free-text description, the location at which the item was lost or found, and an approximate date and time. Each new report is compared against the opposite pool of reports (lost against found, and vice versa) to produce a ranked list of candidate matches. A claimant on a candidate match must then pass an evidence-based verification step before the system recommends releasing the item, with borderline or contested cases routed to an administrator. Figure 1 outlines the five stages of this pipeline.

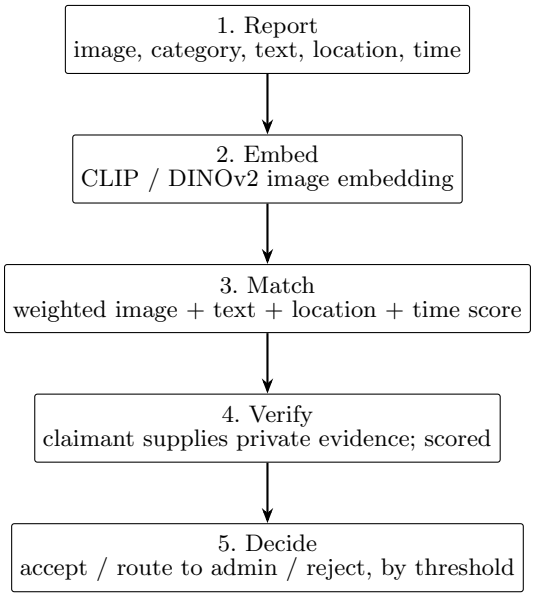


Fig. 1. The five-stage LostLink pipeline, from report submission to a decision on a claim.

4.2 Matching Pipeline

Each item photograph is passed through a pretrained image-embedding model to obtain a fixed-length vector, without any category-specific fine-tuning. Two embedding models are evaluated side by side, CLIP [13] and DINOv2 [11], since the review in Section 2.2 finds no existing lost-and-found study that compares the two directly. Image similarity is computed as the cosine similarity between embedding vectors and combined with text similarity (over the free-text description), location similarity (distance between reported coordinates), and time similarity (closeness of reported date and time) into a single weighted match score:

$$S = w_1 S_{\text{image}} + w_2 S_{\text{text}} + w_3 S_{\text{location}} + w_4 S_{\text{time}}, \qquad \sum_i w_i = 1, \tag{1}$$

following the fusion structure used by Dhanawardhana et al. [8], with the weights w_i tuned on a held-out validation split rather than fixed a priori. Candidate matches are ranked by S and presented to the reporting user together with the four component scores, so that a match can be inspected rather than taken on faith.

4.3 Ownership Verification Mechanism

A high match score establishes that two reports plausibly describe the same item; it does not establish who owns it. When a user submits a claim against a

candidate match, LostLink asks for a small number of private details that were not shown in the public listing, for example a distinguishing mark, the contents of a bag, or proof of purchase. Each answer contributes a weighted point score, and the resulting verification score $V \in [0, 1]$ is compared against an acceptance threshold τ. Where multiple claimants contest the same item, their verification scores are ranked, and the system reports the top-ranked claimant together with the score gap to the next-ranked one, again using identification-rate-at-rank as the evaluation criterion rather than simple top-1 accuracy, following the approach used for multi-candidate identification in [2].

4.4 Dataset Strategy

No public dataset of lost-and-found item pairs with associated ownership claims was located during the review; the systems in Section 2.2 that report strong accuracy each construct their own small, category-specific dataset [8, 19]. LostLink accordingly builds a dataset spanning several everyday item categories, for instance bags, phones, keys, water bottles, and identity cards, collected by photographing the same physical items under varying conditions of lighting, angle, and background to create genuine matching pairs, together with visually similar but distinct items of the same category to create hard negative pairs. Claimant evidence for the verification experiment is collected separately: genuine claimants are shown the true item and asked the verification questions, while simulated fraudulent claimants are shown only the public listing and asked to answer the same questions without access to the private details.

4.5 Evaluation Plan

Matching performance is evaluated using precision, recall, F1-score, and top-K retrieval accuracy, computed separately for CLIP, DINOv2, and a keyword/category baseline, both within a single item category and across categories, to test the cross-category generalisation claim identified as the first gap in Section 3. Verification performance is evaluated using False Acceptance Rate and False Rejection Rate at varying thresholds τ, reported together rather than as a single accuracy figure, in line with the authentication-evaluation practice reviewed in Section 2.4 [1, 2]. Multi-claimant ranking is evaluated by identification-rate-at-rank over the simulated contested-claim cases described above. Table 2 summarises the planned metrics against the corresponding literature precedent.

Table 2. Planned evaluation metrics and their precedent in the reviewed literature.

Component	**Metric(s)**	**Precedent**
Image / fused matching	Precision, recall, F1, top-K accuracy	[8, 19, 12]
Ownership verification	FAR, FRR across thresholds	[1, 2]
Multi-claimant ranking	Identification-rate-at-rank	[2]

Based on the accuracy figures gathered in the review, spanning roughly 75–97% depending on task and category breadth (75% for a single-category fine-tuned Siamese network, 87% for that same system's multimodal fusion, 96.8% for a lightweight image-only matcher tested on one item type, and 95.2% for CLIP on a generic ten-class benchmark [8, 19, 7]), none of which was measured on the specific, cross-category, near-duplicate matching problem LostLink addresses, the working target for the matching pipeline is 90–95% accuracy across multiple item categories. This is stated as a design target to be evaluated once the system is built and tested, not as a result already obtained.

5 Conclusion and Future Work

Lost-and-found recovery has been studied from several angles: keyword and metadata search, category-specific Siamese image matching, general-purpose zero-shot embeddings, multimodal fusion of image with text, location and time, and, least developed of all, ownership verification. Convolutional Siamese networks deliver strong accuracy but do not generalise past the category they were trained on, while zero-shot vision-language and self-supervised models promise to remove that constraint at the cost of being largely untested for this specific application. Multimodal fusion strategies exist and have precedent within lost-and-found systems, but their contribution per modality is rarely isolated and measured. Ownership verification is the least mature area of all: existing systems either skip it, substitute a login check, or describe it conceptually, and none of the reviewed work reports the false-acceptance and false-rejection rates that authentication research treats as the standard way to evaluate exactly this kind of mechanism.

These gaps motivate the proposed methodology in Section 4: LostLink is designed to measure matching accuracy across item categories using zero-shot embeddings and to report FAR/FRR for a scored ownership-verification step, closing both gaps with measurable, reproducible results rather than a further descriptive prototype. Future work, once the system above has been implemented and evaluated, includes extending the item-category set, testing fine-tuning of the embedding models if a sufficiently large in-domain dataset becomes available, and studying whether the explainable score breakdown shown to users changes how administrators resolve contested claims in practice.

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